

```

yuktha@LAPTOP-UAVR6HQ0:~$ cd OSLab
yuktha@LAPTOP-UAVR6HQ0:~/OSLab$ mkdir Skill4
yuktha@LAPTOP-UAVR6HQ0:~/OSLab$ cd Skill4
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Skill4$ gedit shell_parser.c

** (gedit:564): WARNING **: 08:05:23.443: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:564): WARNING **: 08:05:23.443: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Skill4$ gcc shell_parser.c -o shell_parser
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Skill4$ ./shell_parser
Enter command: ls file

----- TOKEN STREAM -----
Token 1: ls                Type: WORD
Token 2: file              Type: WORD

----- PARSE TREE -----
|-- COMMAND (NON_TERMINAL)
|   |-- ls (WORD)
|       |-- file (WORD)

----- EXECUTION STRUCTURE -----
COMMAND/ARGUMENT : ls
COMMAND/ARGUMENT : file

Parsing completed successfully.
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Skill4$ cat file.txt | grep hello
cat: file.txt: No such file or directory
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Skill4$ cat file.txt | grep hello
cat: file.txt: No such file or directory
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Skill4$ ./shell_parser
Enter command: cat file.txt | grep hello

----- TOKEN STREAM -----
Token 1: cat                Type: WORD
Token 2: file              Type: WORD
Token 3: .                  Type: UNKNOWN
Token 4: txt                Type: WORD
Token 5: |                  Type: PIPE
Token 6: grep               Type: WORD
Token 7: hello              Type: WORD

Lexical Error: Unknown token '.'
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Skill4$ ./shell_parser
Enter command: cat file > output

----- TOKEN STREAM -----
Token 1: cat                Type: WORD

```

Enter command: cat file > output

----- TOKEN STREAM -----

Token 1: cat	Type: WORD
Token 2: file	Type: WORD
Token 3: >	Type: OUTPUT_REDIRECT
Token 4: output	Type: WORD

----- PARSE TREE -----

```
|-- COMMAND (NON_TERMINAL)
  |-- cat (WORD)
    |-- file (WORD)
      |-- > (OUTPUT_REDIRECT)
        |-- output (WORD)
```

----- EXECUTION STRUCTURE -----

COMMAND/ARGUMENT : cat
COMMAND/ARGUMENT : file
OUTPUT REDIRECT : >
COMMAND/ARGUMENT : output

Parsing completed successfully.

yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Skill4\$./shell_parser

Enter command: cat < input

----- TOKEN STREAM -----

Token 1: cat	Type: WORD
Token 2: <	Type: INPUT_REDIRECT
Token 3: input	Type: WORD

----- PARSE TREE -----

```
|-- COMMAND (NON_TERMINAL)
  |-- cat (WORD)
    |-- < (INPUT_REDIRECT)
      |-- input (WORD)
```

----- EXECUTION STRUCTURE -----

COMMAND/ARGUMENT : cat
INPUT REDIRECT : <
COMMAND/ARGUMENT : input

Parsing completed successfully.

yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Skill4\$./shell_parser

Enter command:

Empty command entered.

yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Skill4\$ |

```
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Skill4$ ./shell_parser  
Enter command:
```

```
Empty command entered.
```

```
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Skill4$ |
```